Forecasting COP–USD

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Data Statistics

Table: Descriptive statistics of exchange rates

Statistic	BRL/USD	CLP/USD	COP/USD
N	8360	8360	8360
μ	2.6786	601.0631	2397.5545
σ	1.5013	153.7802	998.2891
min	1.0045	368.7500	735.0000
$Q_{0.25}$	1.7335	482.3200	1802.5000
$Q_{0.50}$	2.2716	567.0000	2279.6450
$Q_{0.75}$	3.5974	690.8075	2989.4825
max	6.2931	1048.5000	5120.1500

Data Series

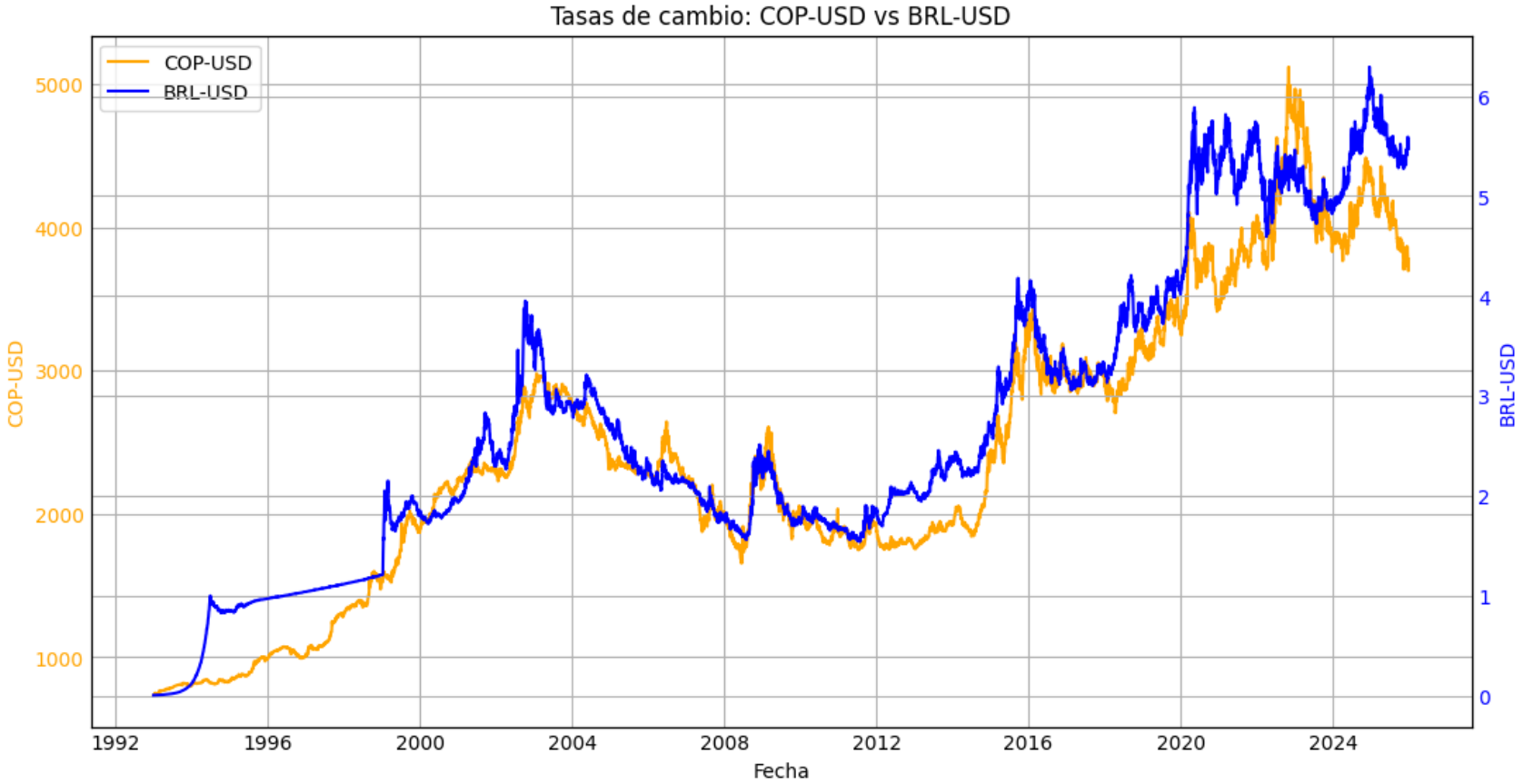

Figure: Time evolution of COP/USD and BRL/USD exchange rates (1993–2025).

Data Series

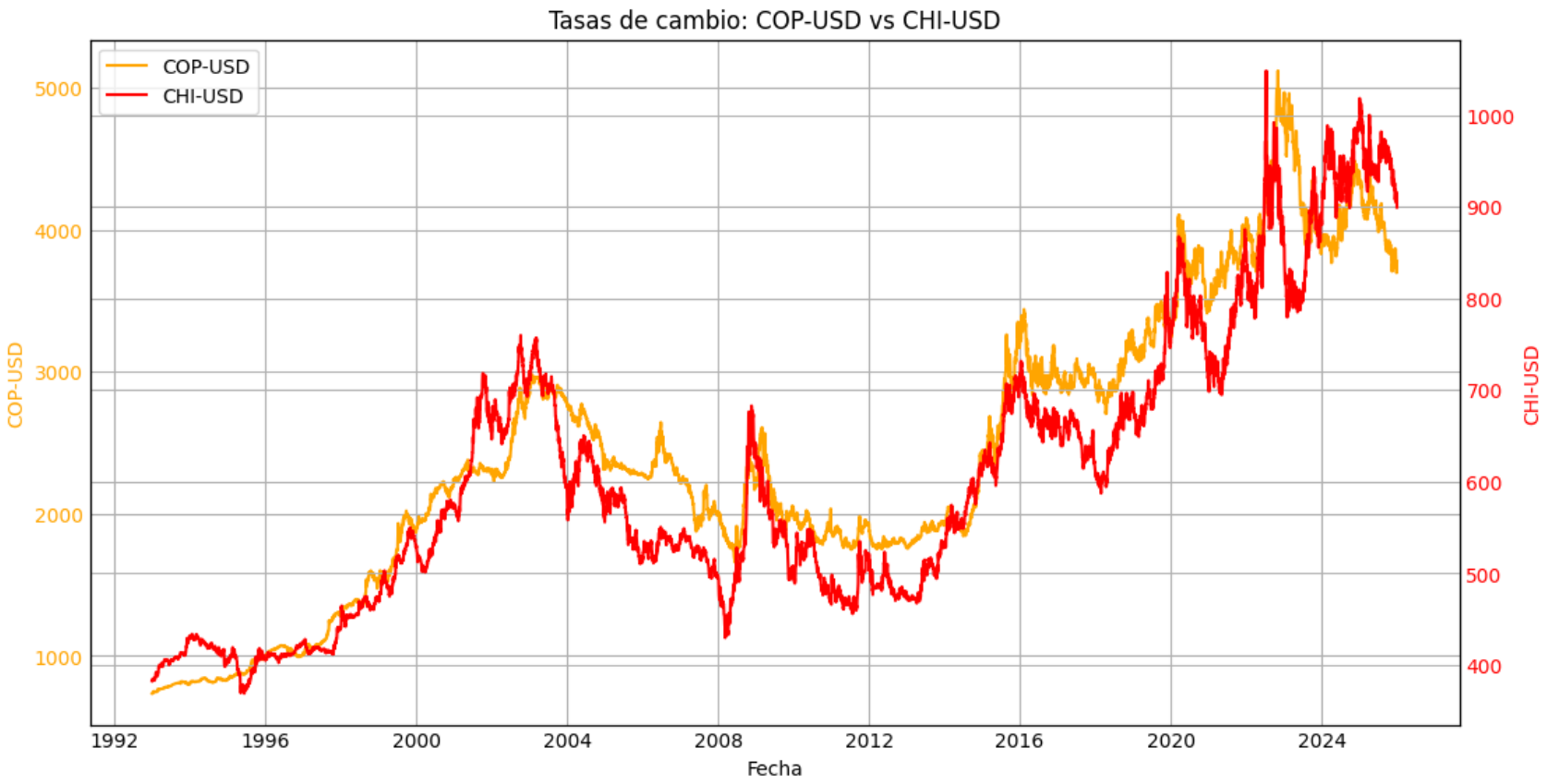

Figure: Time evolution of COP/USD and CLP/USD exchange rates (1993–2025).

(DFGCN) Full Prediction – 1 Week

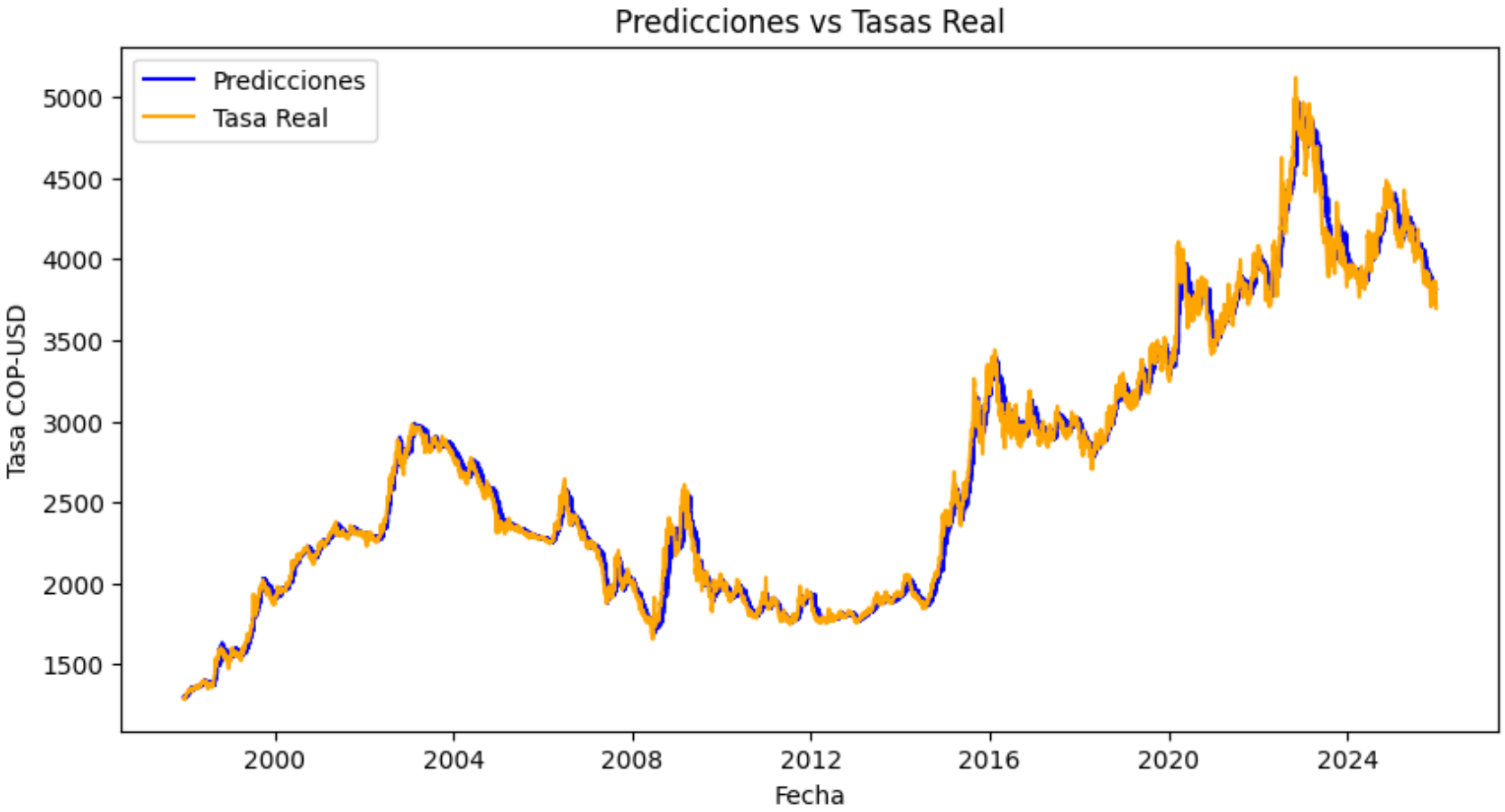

Figure: Time evolution, 1-week horizon, of predicted and actual COP/USD exchange rates (1997–2025).

(DFGCN) Full Prediction – 6 Months

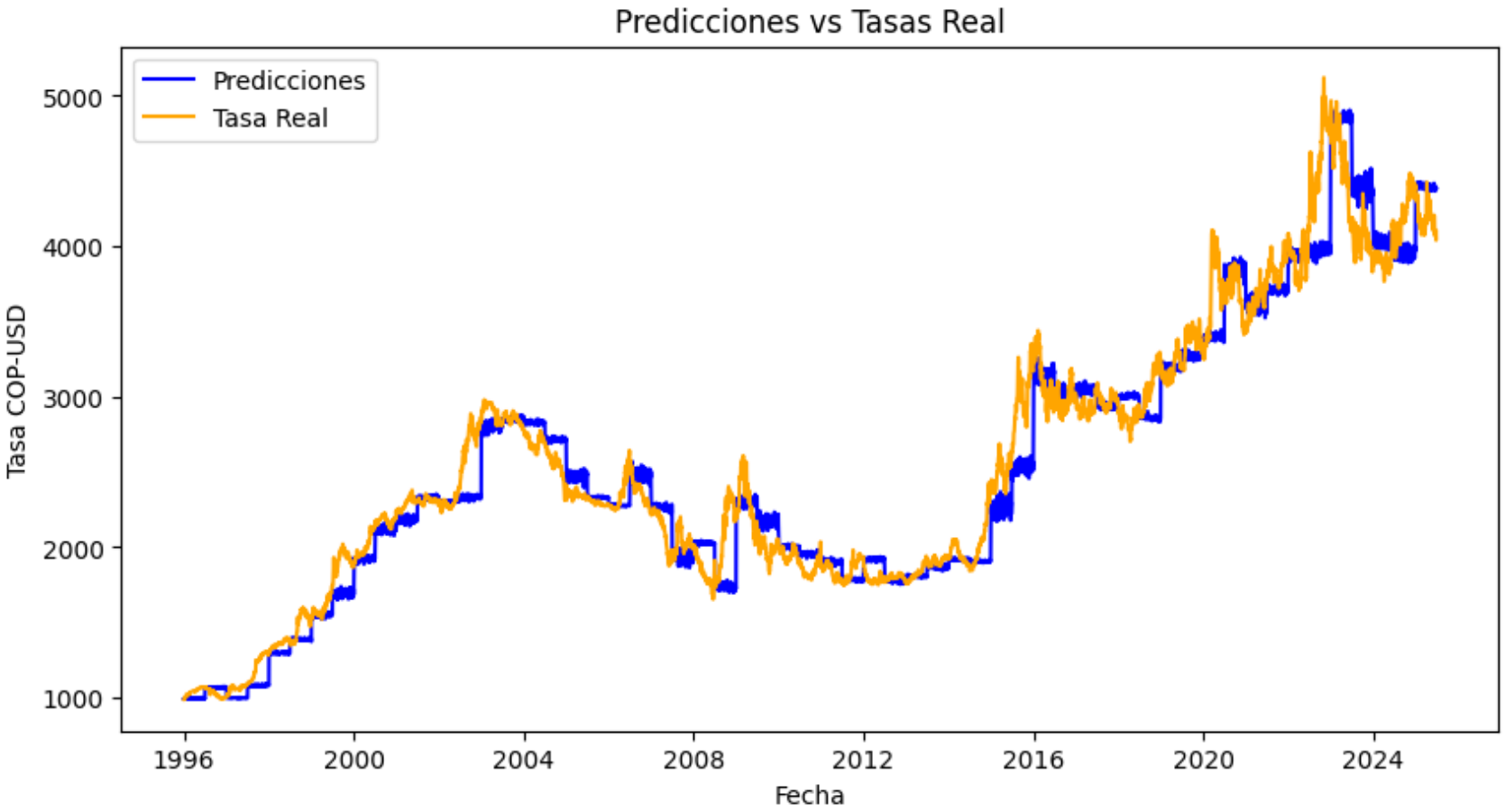

Figure: Time evolution, 6-month horizon, of predicted and actual COP/USD exchange rates (1996–2025).

(DFGCN) Full Prediction – 1 Year

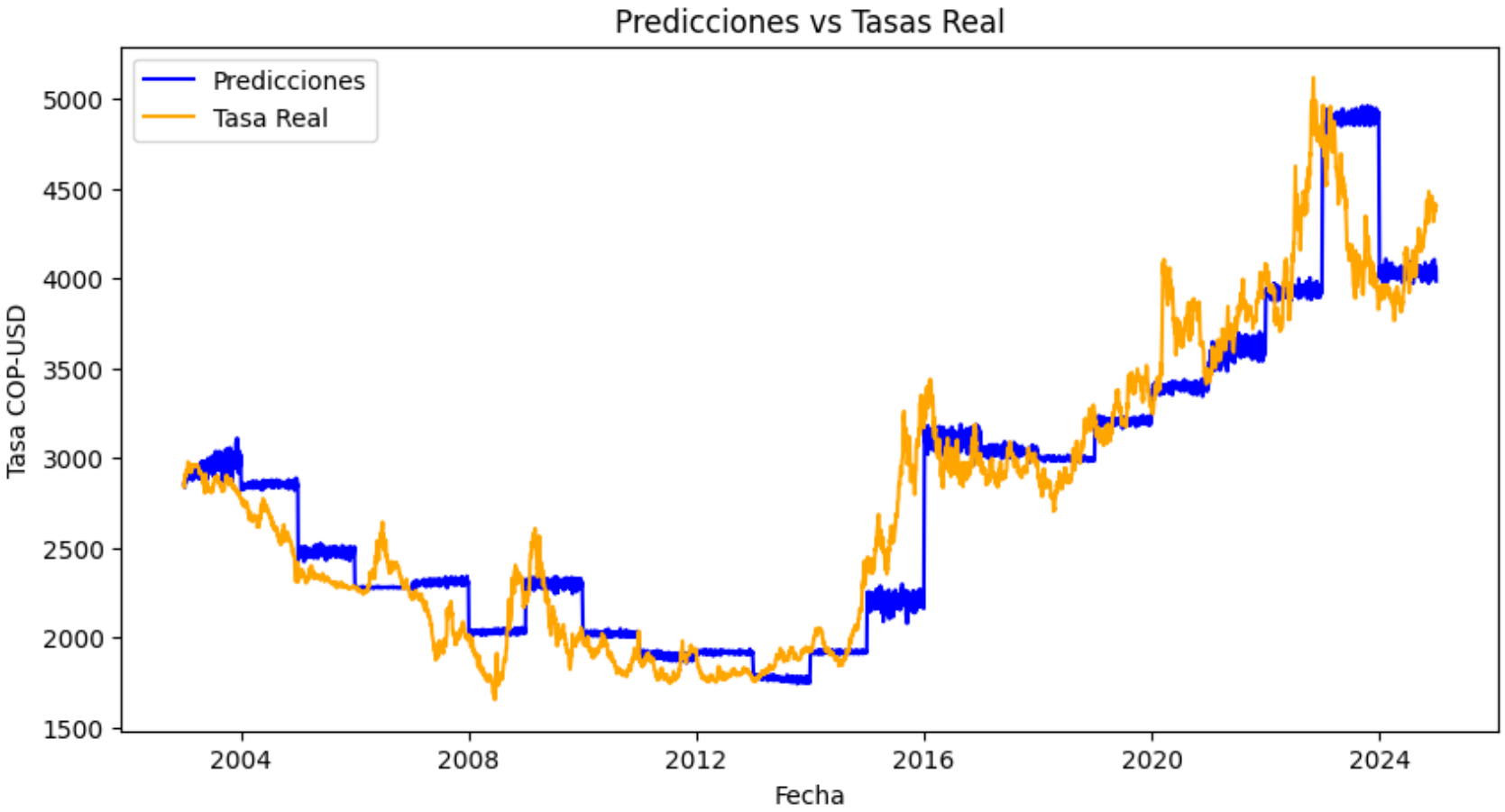

Figure: Time evolution, 1-year horizon, of predicted and actual COP/USD exchange rates (2002–2025).

(DFGCN) Last 5 Years Prediction – 1 Week

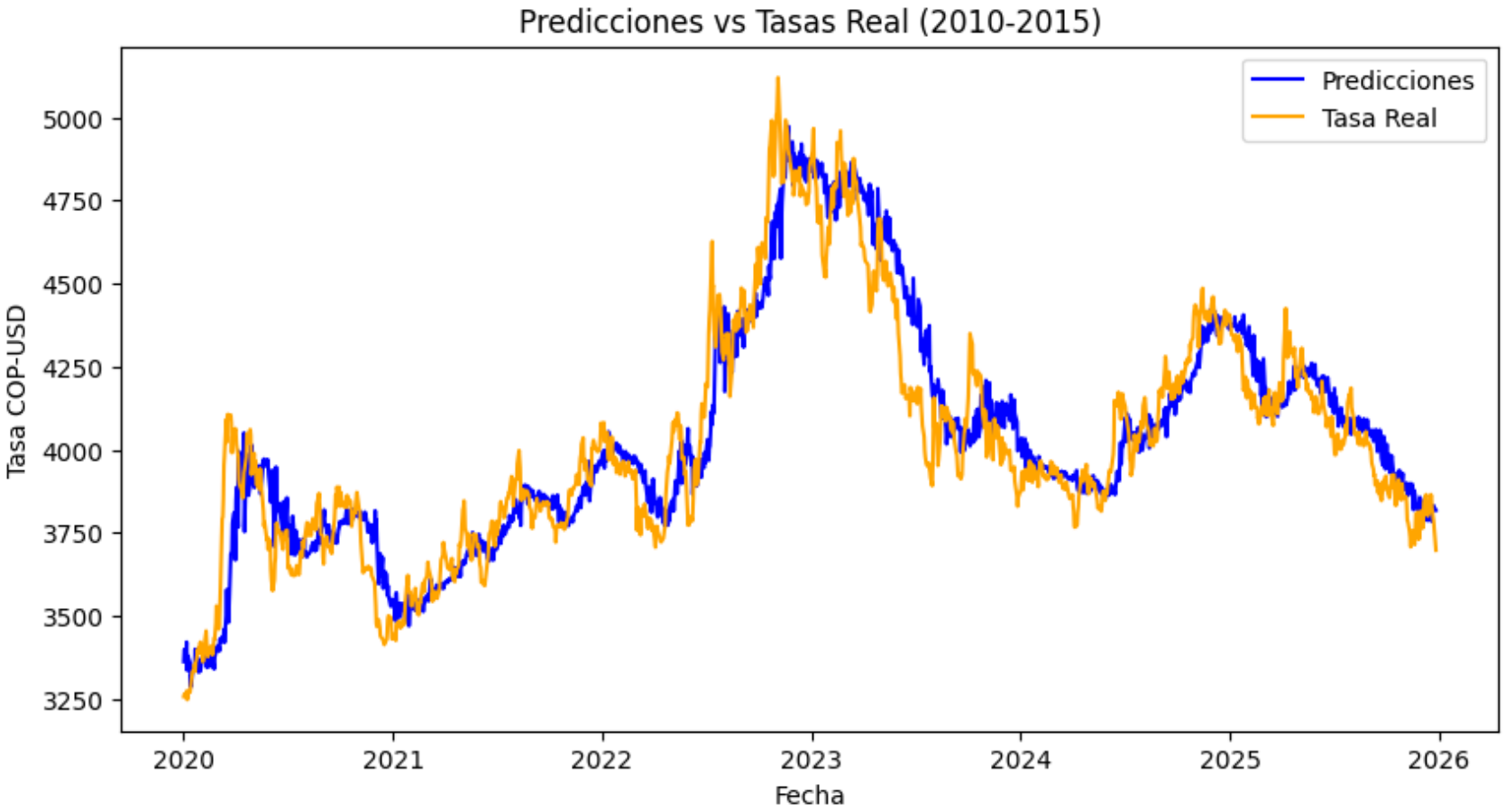

Figure: Time evolution (last 5 years), 1-week horizon, of predicted and actual COP/USD exchange rates (2020–2025).

(DFGCN) Last 5 Years Prediction – 6 Months

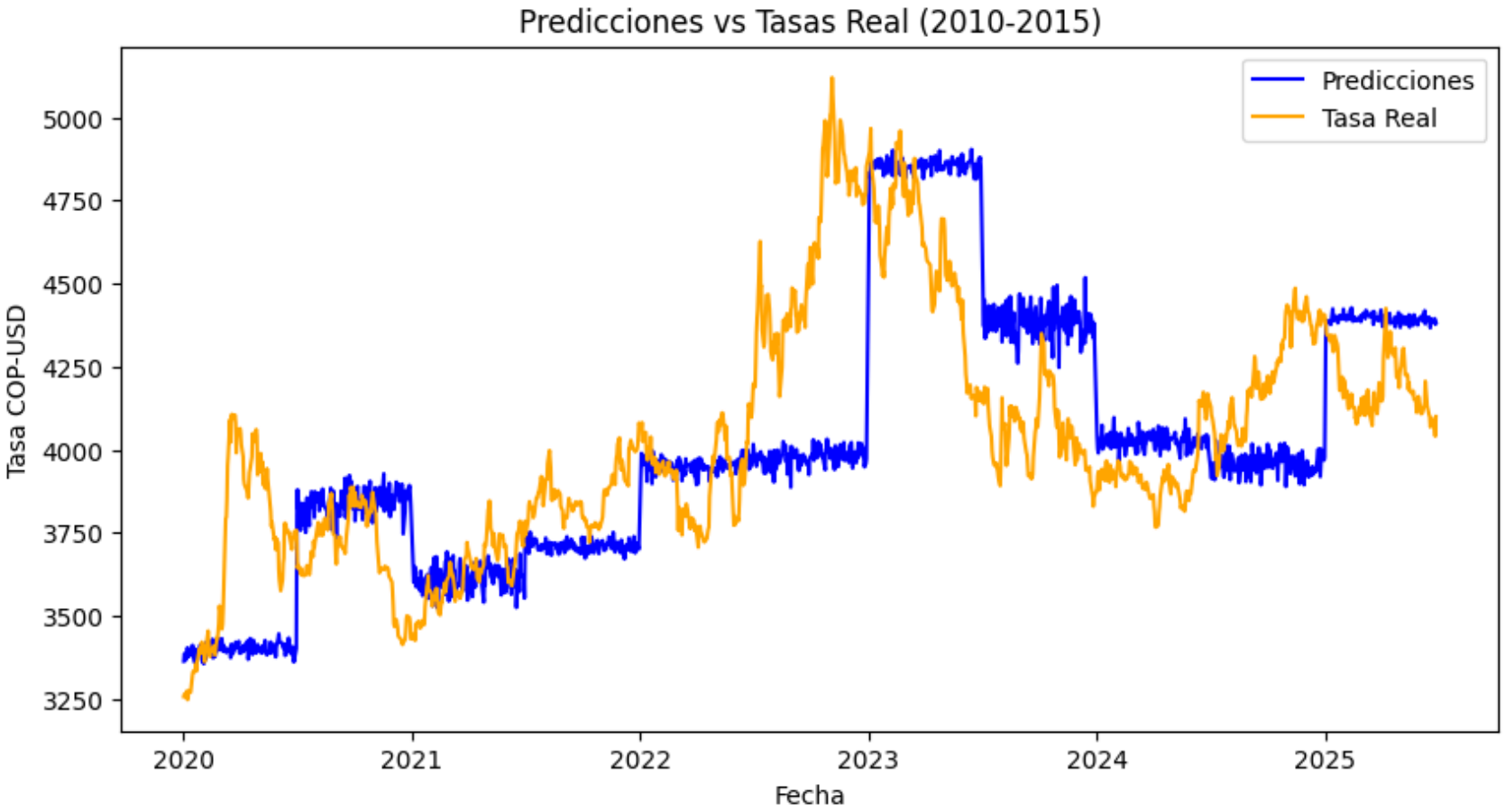

Figure: Time evolution (last 5 years), 6-month horizon, of predicted and actual COP/USD exchange rates (2020–2025).

(DFGCN) Last 5 Years Prediction – 1 Year

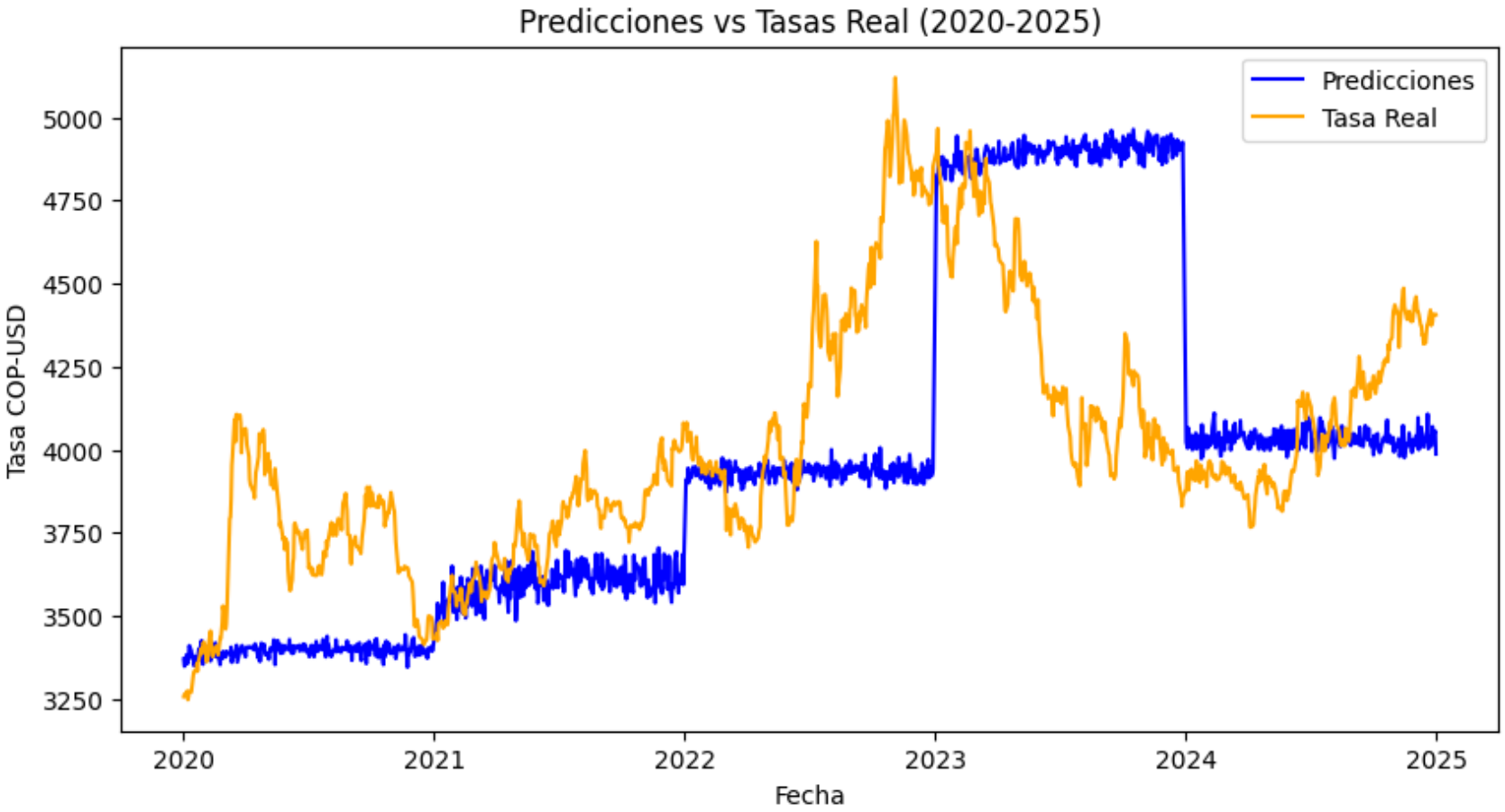

Figure: Time evolution (last 5 years), 1-year horizon, of predicted and actual COP/USD exchange rates (2020–2025).

(RW) Full Prediction – 1 Week

Figure: Time evolution, 1-week horizon, of predicted and actual COP/USD exchange rates (1993–2025).

(RW) Full Prediction – 6 Months

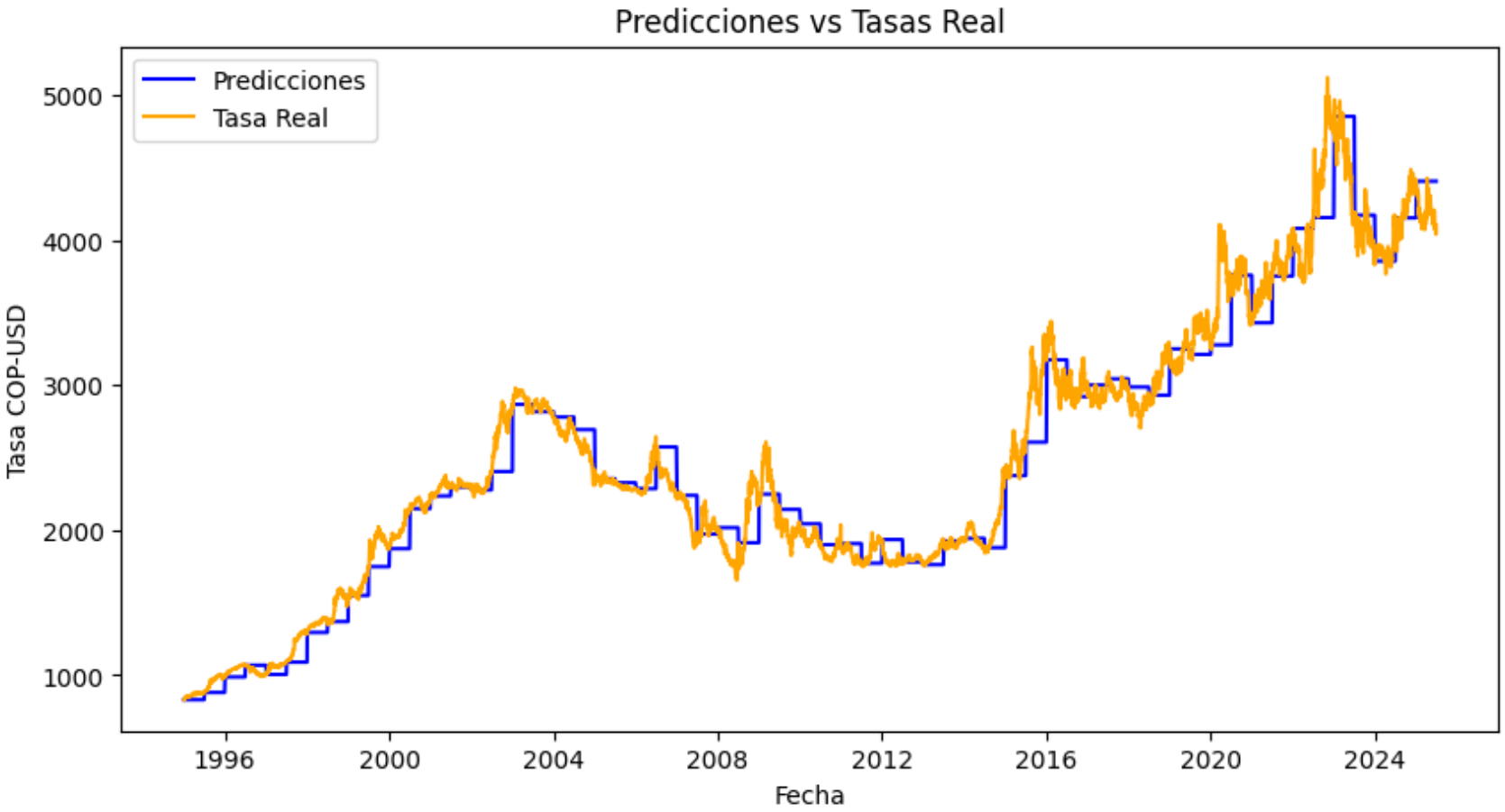

Figure: Time evolution, 6-month horizon, of predicted and actual COP/USD exchange rates (1994–2025).

(RW) Full Prediction – 1 Year

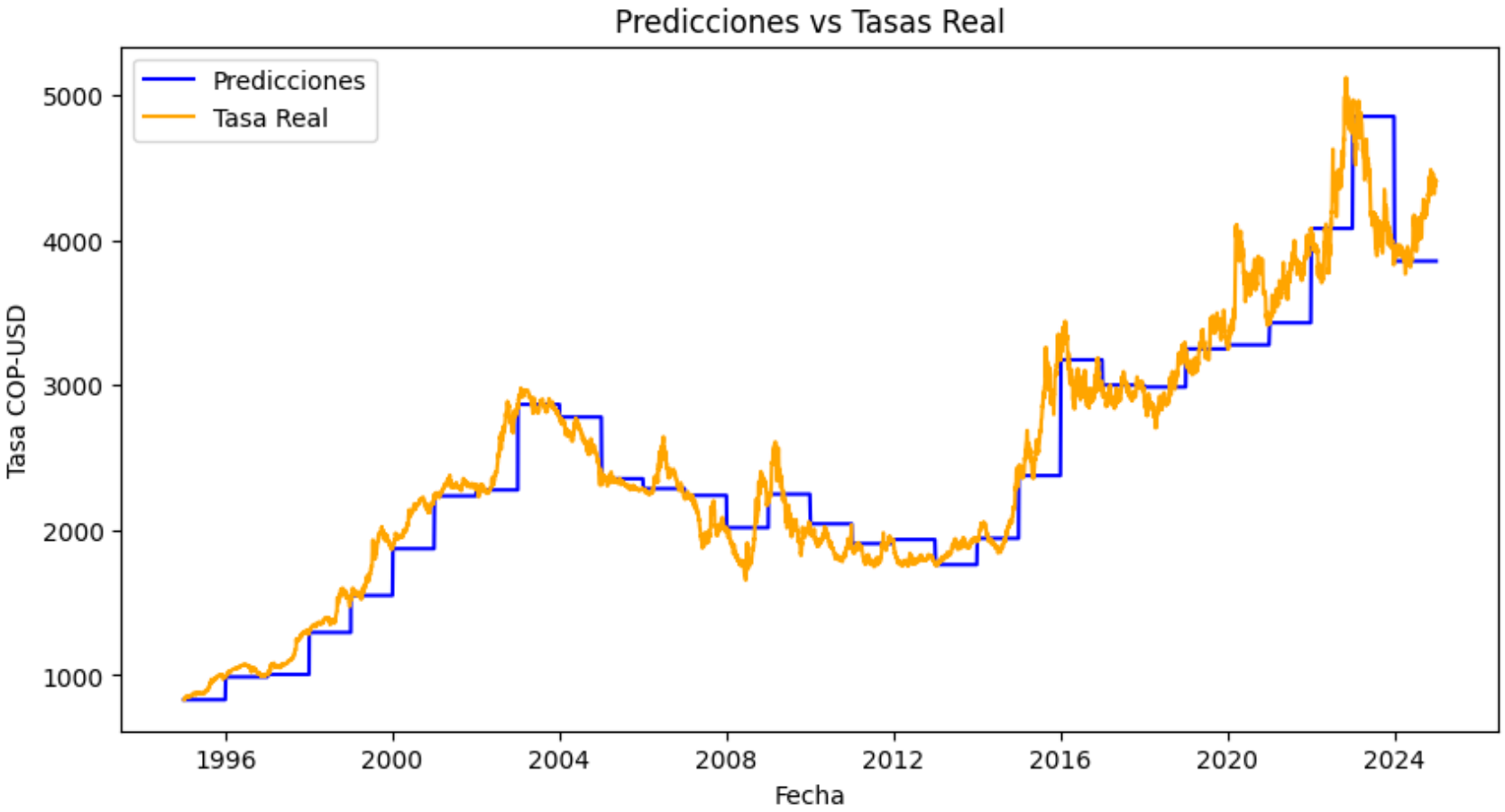

Figure: Time evolution, 1-year horizon, of predicted and actual COP/USD exchange rates (1995–2025).

(RW) Last 5 Years Prediction – 1 Week

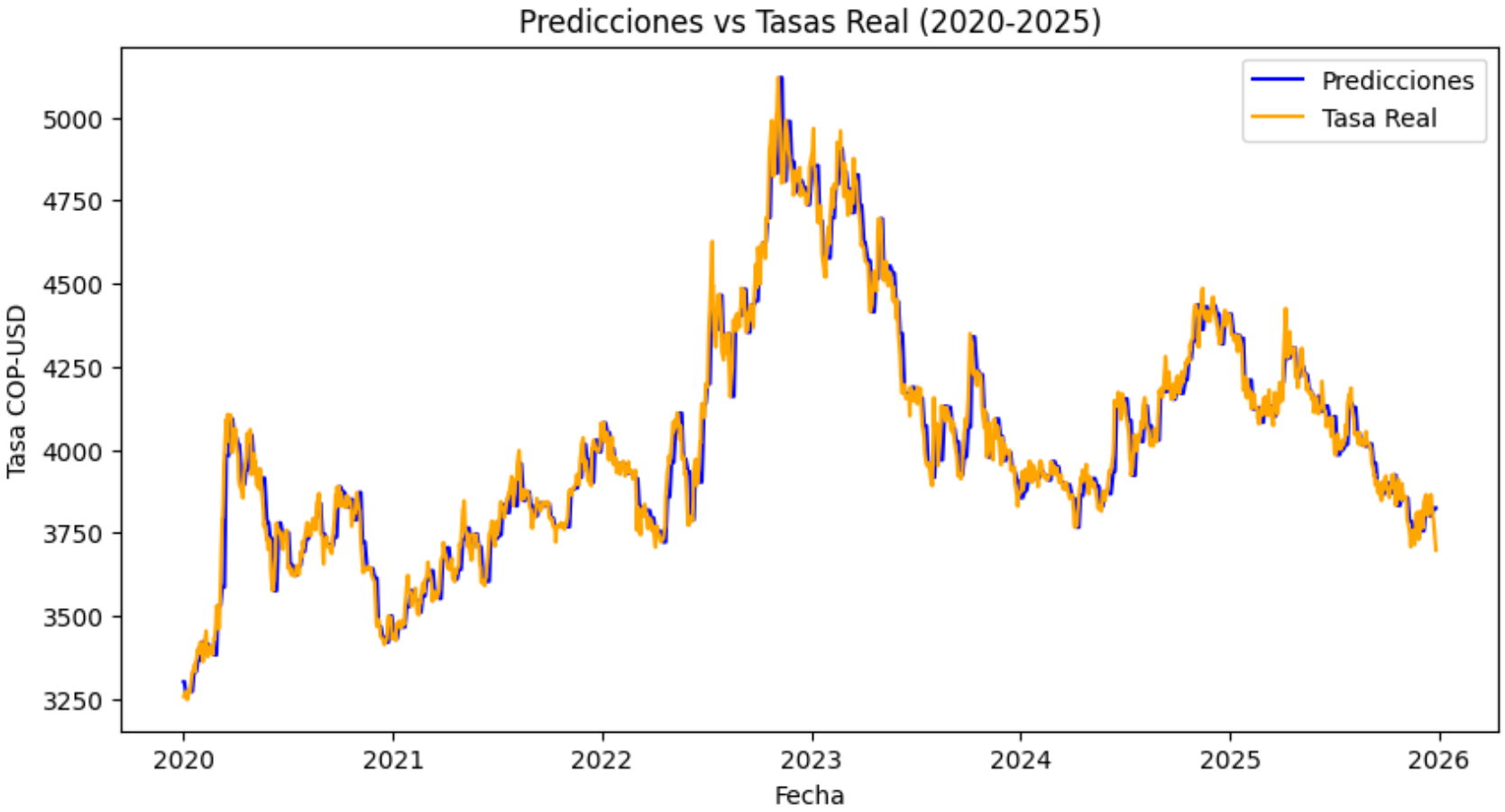

Figure: Time evolution (last 5 years), 1-week horizon, of predicted and actual COP/USD exchange rates (2020–2025).

(RW) Last 5 Years Prediction – 6 Months

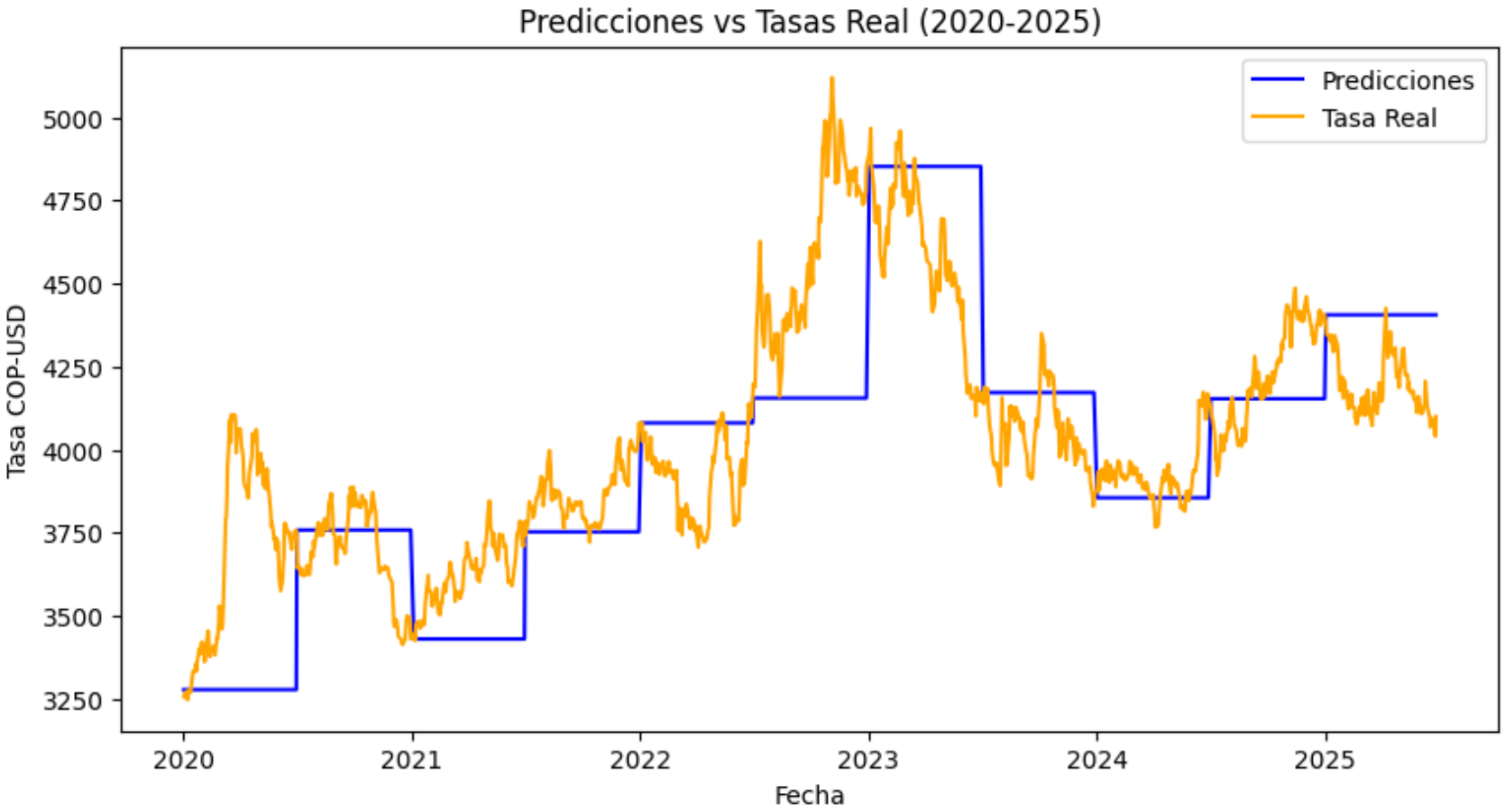

Figure: Time evolution (last 5 years), 6-month horizon, of predicted and actual COP/USD exchange rates (2020–2025).

(RW) Last 5 Years Prediction – 1 Year

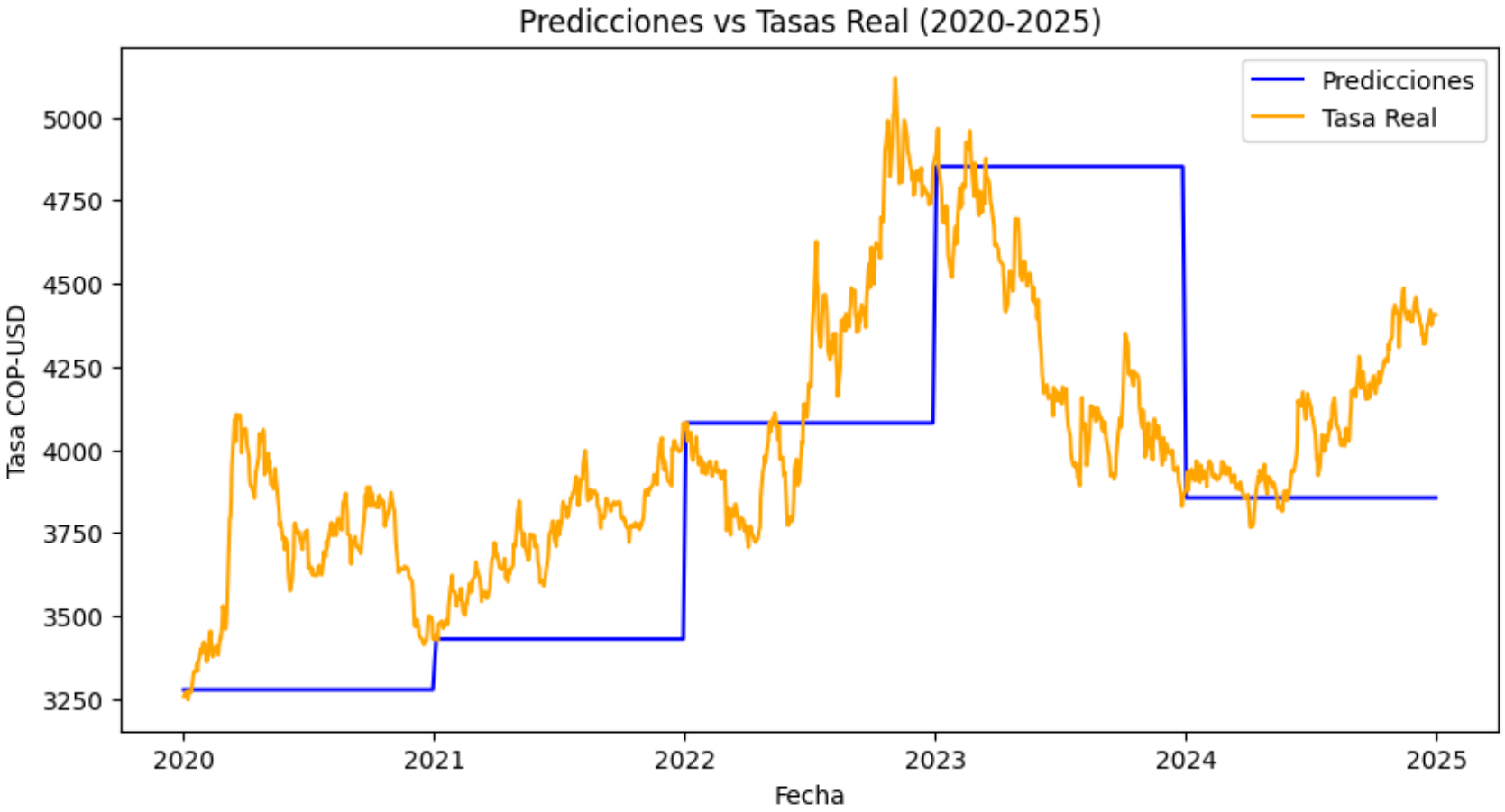

Figure: Time evolution (last 5 years), 1-year horizon, of predicted and actual COP/USD exchange rates (2020–2025).

Resultados DM

Horizon	Train Size	DM	p-value	Accurate	Model	MAE DFGCN	MAE RW
1W	52 W	23.2346	0.0000	True	RW	55.1614	22.2344
1W	104 W	22.4450	0.0000	True	RW	55.5265	22.9130
1W	156 W	22.1492	0.0000	True	RW	55.9776	23.5772
1W	260 W	19.8500	0.0000	True	RW	55.1500	24.9938
2W	52 W	20.8619	0.0000	True	RW	61.1162	29.7533
2W	104 W	19.9429	0.0000	True	RW	61.3145	30.6602
1M	12 M	26.0330	0.0000	True	RW	73.5638	47.5440
1M	24 M	17.3665	0.0000	True	RW	74.7552	48.9794
2M	12 M	13.7725	0.0000	True	RW	89.4926	68.1135
2M	24 M	12.2734	0.0000	True	RW	89.4118	70.1412
3M	24 M	10.4935	0.0000	True	RW	101.7707	85.2823
3M	36 M	10.2420	0.0000	True	RW	104.1762	87.4573
6M	24 M	7.9778	0.0000	True	RW	135.7217	121.8195
6M	36 M	8.4656	0.0000	True	RW	141.6080	124.1599
1Y	3 Y	0.6114	0.5409	False	ND	178.0093	176.8441
1Y	5 Y	1.1546	0.2483	False	ND	186.1144	183.6278
1Y	10 Y	3.1167	0.0018	True	RW	198.0546	186.3763
2Y	6 Y	4.7261	0.0000	True	RW	293.7253	279.0585
2Y	10 Y	4.4875	0.0000	True	RW	308.4477	280.3887

Table: Diebold–Mariano test results